

Shahryar motamedi

astrophysics, galactic dynamics & modified gravity

Nationality: Iranian

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Summary

Computational astrophysicist specializing in large-scale numerical modeling and data analysis using Python in Linux-based high-performance computing (HPC) environments, with hands-on experience in using RAMSES (from execution to analysis) as a high resolution open source code for hydrodynamical and gravitational N-body astrophysical simulations, to probe fundamental unsolved questions in galactic dynamics and modified gravity theories.

Online Profiles

GitHub — github.com/shmotamedi

ResearchGate — researchgate.net/profile/Shahryar-Motamedi

LinkedIn — linkedin.com/in/shahryar-motamedi-26b169b7

Education

Ph.D. in Astrophysics

2015–2024

Institute for Advanced Studies in Basic Sciences (IASBS), Zanzan, Iran

Advisor: Prof. Hosein Haghi (haghi@iasbs.ac.ir)

M.Sc. in Space Engineering

2010–2013

Sharif University of Technology (SUT), Tehran, Iran

Advisor: Prof. S. H. Pourtakdoust (pourtak@sharif.edu)

B.Sc. in Physics

2006–2010

Sharif University of Technology (SUT), Tehran, Iran

Research Background

Developing a new approach to Alfvén wave turbulence

Research Assistant

IASBS, Zanzan, Iran

Comparing dynamical-friction timescales in Milgromian and Newtonian dynamics

Research Assistant

SUT, Tehran, Iran

Investigating chaos in the relativistic restricted three-body problem

Visiting Researcher

USTC, Hefei, China (2019)

Developing comparative methods between MOND and DM-based frameworks

Predoctoral Researcher

RCISP, Tehran, Iran (2014)

Designing high-precision orbit propagators using analytical perturbation theory

Selected Publication

Motamedi, S., Haghi, H., Ghari, A., Wu, X. and Kroupa, P. (2026). Dynamical friction in Newtonian and Milgromian dynamics. Monthly Notices of the Royal Astronomical Society (MNRAS).

Honors & Awards

Ph.D. Entrance Exam — Ranked first in IASBS entrance exam

M.Sc. Admission — Awarded full scholarship at SUT with direct admission